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Towards a wildfire vulnerability index using expert judgement

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ABSTRACT

Background. Climate change is expected to significantly change wildfire frequency and magnitude, which poses particular challenges for countries with limited experience in managing wildfires, such as Austria. **Aims.** To develop a vulnerability index, this study aims at weighting the characteristics of buildings and their surroundings (vulnerability indicators) by involving different national stakeholders (Austria) and international experts. **Methods.** Expert judgement and the analytic hierarchy process (AHP) were used for weighting indicators identified in a previous study through a literature review. **Key results.** The judgement of two expert groups regarding the weighting of the vulnerability indicators is compared. A wildfire vulnerability index that combines all the indicators into a single vulnerability index assigned to each building is introduced. **Conclusions.** The wildfire vulnerability index for buildings is a tool for decision-makers and other end users in Austria. The index sets the foundations for ongoing research in the field of vulnerability assessment and is based on trans-disciplinary approaches involving both academia and stakeholders. **Implications.** The index can be used to support decision-making, risk reduction and climate change adaptation strategies but it can also guide local adaptation at the building level.

Keywords: analytic hierarchy process, Austria, buildings, expert judgement, index, indicators, stakeholders, vulnerability, wildland–urban interface, wildfire.

Introduction

Climatic conditions favourable to wildfires are increasing worldwide, including in Europe, but trends in the number of fires and the area burnt vary regionally (Jones *et al.* 2022). Additionally, according to Cunningham *et al.* (2024), the frequency of energetically extreme events associated with extreme environmental, social and economic impacts has increased by a factor of 2.2 over the past 20 years. The global costs of wildfires have risen substantially (Ritchie *et al.* 2022). Moreover, countries with limited experience in managing wildfires in northern and central Europe are expected to experience more wildfires in the future (Cardil *et al.* 2021). Specifically in Austria, which is the focus of the present study, the effects of climate change are expected to lead to an increase in heat waves and a decrease in precipitation (Olefs *et al.* 2021) and, consequently, an increase in fire weather days, which will ultimately increase wildfire risk in the country (Müller *et al.* 2020).

Increased wildfire risk is influenced not only by climate change but also by socio-economic dynamics. Austrian forests, for example, serve as recreational areas and as a source of timber as well as protection against other natural hazards such as snow avalanches and landslides. Until now, wildland fires in Austria have had only a minor impact on human lives, settlements and infrastructure within the wildland-urban interface (WUI). However, focusing on the WUI is important as 85% of forest fires in Austria have been caused directly or indirectly by human activity, and the remaining 15% by lightning (Müller *et al.* 2020). The interaction between forests and settlements is twofold, as on the one hand, human activities increase the likelihood of wildfires, but on the other hand, wildfires threaten the population and infrastructure in the WUI. It is, therefore,

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pivotal to focus on the vulnerabilities within the WUI that affect overall wildfire risk to our society.

Studies focusing on the vulnerability of the built environment do exist, but they are mostly limited to the interaction between buildings and fire, rather than providing a tool to measure this vulnerability for risk reduction purposes. However, some of them focus on assessing the vulnerability of buildings within the WUI in Europe. These studies include the study by Samora-Arvela *et al.* (2023), who investigated the factors that contribute to the resistance of buildings to the impact of wildfire to support municipal mitigation strategies in Portugal. Moreover, the relationship between building characteristics and the resulting wildfire damage to buildings was investigated by Dossi *et al.* (2023) using data from California and Portugal. In terms of expert judgement, Àgueda *et al.* (2023) developed a tool for homeowners based on fuzzy logic (fuzzy preferences of WUI experts) to help them assess and reduce the physical vulnerability of buildings to wildfire, which was applied in two case studies in Spain. One of the first attempts to develop a wildfire vulnerability index was the study by Papathoma-Köhle *et al.* (2022*b*), based on building characteristics and damage data from the wildfire in Mati, Greece (July 2018).

Nevertheless, all these studies use damage data from countries with long experience of wildfire and a large amount of empirical data. The lack of empirical data in countries with a poor record of wildfire events is challenging for vulnerability and risk assessment. A promising approach to overcome this gap is to compensate for the lack of data with expert judgement or other participatory methods.

In this paper, we focus on such an approach. By examining buildings located within the WUI and considering their characteristics along with their surroundings, we have developed a vulnerability index for these structures. This index can guide disaster risk reduction strategies, building reinforcement efforts and preparedness policies for future wildfire events. The outcomes of our approach are essential for reducing the vulnerability and increasing the resilience of the built environment. Herein, 'vulnerability' is considered to be the conditions determined by physical (in this case) factors that increase the susceptibility of an asset to the impacts of hazards, and 'resilience' is considered to be the capacity to resist, absorb, accommodate, adapt to, transform and recover from these impacts (UNISDR 2009).

Wildfire risk in Austria and research needs

In Austria, wildfires are not as frequent as in other European countries (e.g. Portugal, Greece); nevertheless, when they do occur, they can have costly consequences. For example, a wildfire in the Austrian Federal State of Tyrol in 2014 resulted in costs of ~€3 million (Müller *et al.* 2020). In 2014 and 2015, Austria experienced two large wildfires, each burning approximately 100 ha. In 2021, a wildfire in the Austrian Federal State of Lower Austria (Hirschwang an der Rax) was recorded as one of the largest wildfires ever in Austria, burning more than 115 ha. Nevertheless, in the following year (2022) the burnt area in Austria increased dramatically mainly owing to a single event (Allensteig, Lower Austria) that was responsible for burning more than 800 ha (including 550 ha of forest) (Müller 2023). To monitor wildfires in Austria, an online database (Vacik and Müller 2013) with more than 7000 data records of current and historical forest fires has been established. Based on this database, wildfire events in Austria since 2000 are shown in Fig. 1. Most wildfires (85%) were caused by humans, with tourism and recreational activities being the most important factors for their initiation (Müller *et al.* 2020).

In recent years, significant effects of climate change have become manifest in the greater Alpine region, leading to severe precipitation deficits in Austria (Haslinger and Mayer 2023). Long intense dry periods in 2011 and 2012 led to severe spring fires, whereas summer fires increased in 2013 (Fig. 1). In 2015, an extremely dry late autumn resulted in wildfires that lasted until the end of December in the high mountains. Rising temperatures and the resulting dry conditions caused by human-induced climate change have repeatedly been blamed for these anomalies (Vacik *et al.* 2020). The changes observed in Austria have also been recorded at

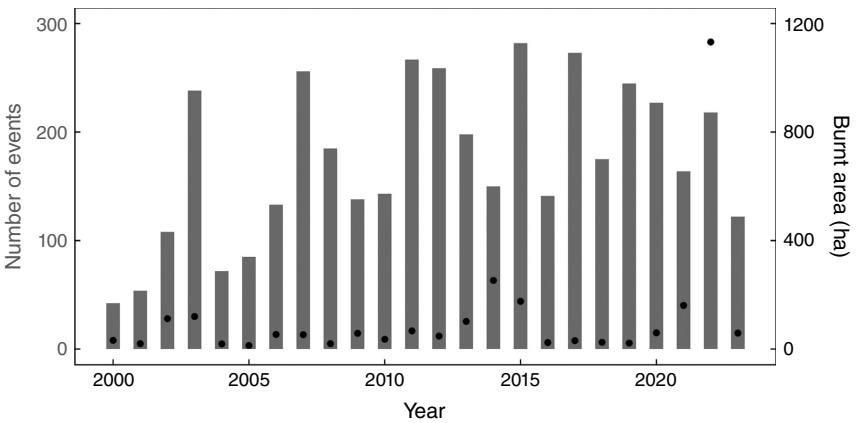


Fig. 1. Annual number of wildfires (bars) and burnt area (dots) in Austria since 2000 (data source: https://fire.boku.ac.at).

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the European level by EFFIS (European Forest Fire Information System). Based on these data and according to the European Commission: Joint Research Centre *et al.* (2022), there is an increasing trend towards a higher number and larger area of forest fires in the European Union (EU). As wildfires occur in Austria and other regions with limited historical records of such events (e.g. Scandinavia, Siberia), central and northern European countries are seeking ways to reduce wildfire risk. In Austria, although efforts have been made to assess wildfire risk, most of them do not include a vulnerability study. Current efforts are focusing on hazard assessment (Müller *et al.* 2020), which is an essential component for the assessment of associated risks, but it needs to be complemented by vulnerability studies.

Method

As empirical data on wildfire losses are missing, the methodological approach used herein relied on expert judgment to determine the physical vulnerability of buildings located in the WUI. The workflow included the following steps: (a) a thorough literature review on vulnerability indicators of buildings exposed to wildfire; (b) stakeholder mapping to identify Austrian and international experts able to assess the importance of these indicators; (c) the identification of relevant vulnerability indicators and the design of a questionnaire to assess these indicators; (d) the implementation of the analytic hierarchy process (AHP) to weight the indicators; and (e) the development of a wildfire vulnerability index. The workflow of the study is shown in Fig. 2.

Literature review

In the literature review that was completed during a previous study (Papathoma-Köhle *et al.* 2022*b*), 35 peer-reviewed scientific articles published over approximately the last 30 years (1991–2020) were examined. Their focus was buildings and their interaction with wildfire in the WUI in case studies in Europe, the USA and Australia (e.g. Xanthopoulos 2004; Australian Standards 2009; Lampin-Maillet *et al.* 2010; Maranghides *et al.* 2013; Quarles *et al.* 2013; Laranjeira and Cruz 2014; Alexandre *et al.* 2016*a*; Syphard and Keeley 2019; Vacca *et al.* 2020). Although vulnerability was not always explicitly addressed in the papers, information was provided on the characteristics of buildings and their surroundings and how these contribute to overall wildfire vulnerability, allowing a list of potential indicators to be compiled (Papathoma-Köhle *et al.* 2022*b*).

Stakeholder mapping

Two main pools of experts were invited as the basis for expert judgement: Austrian stakeholders and international experts. These two groups have different strengths and advantages. On the one hand, the Austrian stakeholders are familiar with the Austrian built environment, but, on the other hand, the international experts bring experience and scientific knowledge from many places in the world. Fourteen Austrian stakeholders were identified as experts on the basis of their involvement in research, decision making or response to wildfires. Their professional backgrounds included research related to forest fires, engineering, natural hazard insurance, emergency services (firefighters) and public administration related to natural hazards in Austria (Table 1). Nine out of the fourteen Austrian stakeholders responded and sent a completed questionnaire, resulting in a response rate of approximately 64%.

More than 50 (54) international experts were invited to provide expert judgement on the project. The main criterion for their selection was their involvement in research related to wildfire impacts in the WUI, as evidenced by their scientific publications over the last 20 years. Consequently, they all had a similar professional background (university, research institute), but with a wide geographical

Table 1. Austrian stakeholders identified as experts on the basis of their involvement in research, decision making or response to wildfires.

Professional background	Invited (14)	Responded (9)
University and research	4	2
Authorities	3	1
Fire brigade	6	5
Insurance	1	1

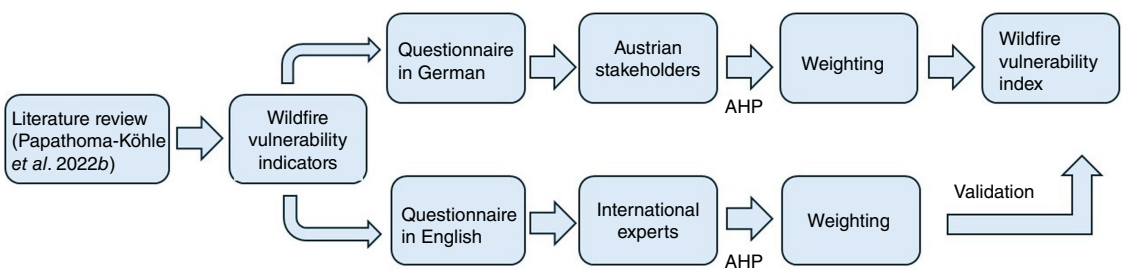


Fig. 2. The methodological approach used in this study.

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Table 2. International stakeholders identified as experts on the basis of their involvement in research, defined by having published on the topic over the past 20 years.

Country	Invited (54)	Responded (19)
USA	21	6
Portugal	5	4
Spain	5	2
Greece	10	5
Australia	6	1
France	4	0
UK	2	1
Canada	1	0

distribution (Table 2), in order to benefit from their experience and knowledge from countries with experience in wildfire management. Of the 54 international experts invited to participate in the survey, 19 completed the questionnaire (35% response rate).

Design of questionnaire

A questionnaire was designed (see Supplementary Appendix S1) on the basis of the literature review and subsequently sent to the two expert groups described in the previous section. This questionnaire was used to weight the different indicators against each other in order to obtain a ranking of the important variables that determine the vulnerability of buildings to wildfire. The set of indicators was divided into the following three categories:

1. Building environment: characteristics of nearby forest including information on the settlement the building is located in and nearby forest (e.g. vegetation density and type).
2. Building exterior: characteristics of the immediate surroundings of the building including ground cover and vegetation.
3. Building characteristics: characteristics of the building including materials, design, shape, etc.

All the indicators considered in the study can be seen in Table 3. It is clear to us that some indicators, especially those in the 'building environment' category, are related to the hazard and less to the element at risk. Nevertheless, as physical vulnerability is hazard-specific, characteristics of the hazard that are related to its impact on buildings are also included in the assessment.

Application of analytic hierarchy process (AHP)

The AHP was used to analyse the results of the questionnaires sent out to the stakeholders. The AHP is part of a prescriptive decision theory and is used to solve multi-criteria decision problems (Saaty 1987). AHP can break down complex decisions into hierarchical structures providing a quantitative basis to support decision-making and, for this reason, is widely used for evaluating expert judgement in various scientific fields. The procedure of the AHP follows the publications of Saaty (1987), Riedl (2005) and Ronninger (2019). The steps of the AHPs are described in detail below:

1. Questionnaire and pairwise comparisons: based on the literature review, a questionnaire was designed. The questionnaire was used to perform indicator weighting based on expert assessment by leveraging the AHP methodology in order to account for the different effects on the vulnerability of a building to wildfires. Starting with the list of indicators identified in three categories of 'building environment' (six indicators), 'building exterior' (seven indicators) and 'building characteristics' (eight indicators) as described above, a survey (questionnaire) featuring a total of $\binom{6}{2} + \binom{7}{2} + \binom{8}{2} = 64$ pairwise comparisons was designed. The number of indicators per group was chosen in accordance with Saaty and Ozdemir (2003) and Ozdemir (2005), i.e. limiting the number of elements in a group to seven plus or minus two. The AHP method is based on quantifying the weights of decision criteria by leveraging an exhaustive set of pairwise comparisons, where the relative importance between indicators is estimated by gathering information on the direction and magnitude of relative preference between each pair. Thus, experts estimate the relative importance of indicators on the physical vulnerability of a building for each pair of indicators across all three categories. Instead of the fundamental 1–9 Saaty scale, we opted to use a reduced four-point ratio scale coded as: 1, equal importance; 2, slight importance; 3, moderate importance; and 4, strong importance. If two indicators are considered to exert identical influence on wildfire vulnerability, the middle checkbox stating 'equal importance' can be selected in the questionnaire. However, if the subjective assessment is associated with a tendency towards one of the two opposite indicators, there is a three-level classification (slight, moderate, strong) for each of the two indicators in which the influence can be selected. The questionnaire was designed as a fillable PDF form structured into the three categories described above to make it easily accessible. PDF form data of the responses were dumped to csv-files using pdftk (Steward 2004) for further processing in R (R Core Team 2024). The three categories were not separately weighted.

2. Derivation of weights: aggregated preference weights were computed by analysing the pairwise comparison

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Table 3. Weights of the vulnerability indicators given by the Austrian stakeholders (Weight (AT)) and the international experts (Weight (INT)) and their difference (Delta).

Category	Indicator	Abbreviation	Weight (AT)	Weight (INT)	Delta
Building environment	Protective strips/firebreaks	Firebreaks	0.2858	0.1979	0.088
Building environment	Vegetation density	Vegetype	0.1705	0.1861	-0.0156
Building environment	Distance to neighbouring buildings	Neighbours	0.1632	0.1335	0.0297
Building environment	Forest type	Forest	0.1543	0.1696	-0.0153
Building environment	Water sources	Water	0.1269	0.0912	0.0357
Building environment	Building density	Density	0.0993	0.1099	-0.0106
Building exterior	Distance of tree crowns/vegetation	Vegdist	0.2579	0.2118	0.0461
Building exterior	Distance to forest edge	Forestdist	0.2443	0.1629	0.0815
Building exterior	Type of vegetation	Vegproperty	0.1349	0.1411	-0.0062
Building exterior	Combustible materials and objects	Combustibles	0.1252	0.1241	0.0011
Building exterior	Terrain slope	Slope	0.094	0.1076	-0.0137
Building exterior	Ground covering	Cover	0.0761	0.0811	-0.005
Building exterior	Property boundary	Boundary	0.0676	0.0752	-0.0076
Building characteristics	Roof material	Roofmat	0.2032	0.1636	0.0396
Building characteristics	Façade	Facade	0.1929	0.1229	0.0699
Building characteristics	Building material	Structure	0.1336	0.1319	0.0018
Building characteristics	Roof overhang	Roofoverhang	0.1217	0.1077	0.0141
Building characteristics	Door/window material	Doorwin	0.1003	0.1233	-0.0229
Building characteristics	Shutters	Shutters	0.0939	0.1196	-0.0257
Building characteristics	Roof shape	Roofcomplex	0.0833	0.1086	-0.0253
Building characteristics	Number of floors	Stories	0.0711	0.038	0.0331

matrices of all responses $k = \{1, \ldots, m\}$ and for each of the three categories $c = \{1, 2, 3\}$. m denotes the total number of responses (9 and 19 for the national and international group, respectively). Each of these pairwise comparison matrices A_k^c is a positive reciprocal matrix of order n that is defined as:

$$A_k^c = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,1} & \cdots & a_{n,n} \end{bmatrix} = \begin{bmatrix} 1 & \frac{w_1}{w_2} & \cdots & \frac{w_1}{w_n} \\ \frac{w_2}{w_1} & 1 & \cdots & \frac{w_2}{w_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{w_n}{w_1} & \frac{w_n}{w_2} & \cdots & 1 \end{bmatrix}$$

where $a_{i,j} > 0$ and $a_{j,i} = \frac{1}{a_{i,j}} \; \forall \; i, j$. $a_{i,j}$ represents the pairwise comparison between the attributes i and j by means of their weights, i.e. $a_{i,j} = \frac{w_i}{w_j}$. Thus, $a_{i,j}$ takes the integer value of the ratio scale if i is more important than j, and its reciprocal if j is more important than i. n is the number of indicators per category, and thus takes the values 6, 7 and 8 in the present case.

First, the individual priorities of each expert are derived for each attribute by computing the geometric average of the normalised preference weights per row. Subsequently, these individual priority weights are aggregated to total weights for each attribute, using the geometric mean as an aggregation function. The geometric mean is used as an aggregation function as it satisfies the reciprocal property of pairwise comparisons, i.e. $a_{j,i} = 1/a_{i,j}$ when aggregating individual judgements to group judgements (Aczél and Saaty 1983; Grošelj and Dolinar 2023). The R package ahpsurvey (Cho 2019) was used for conducting the AHP analysis.

Index development

Several studies use indicators to assess physical vulnerability (Barroca *et al.* 2006; Dall'Osso *et al.* 2009; Agliata *et al.* 2021; Papathoma-Köhle *et al.* 2022*a*). The PTVA (Papathoma Tsunami Vulnerability Assessment) method (Papathoma and Dominey-Howes 2003; Papathoma *et al.* 2003; Dominey-Howes and Papathoma 2007; Dall'Osso *et al.* 2009, 2016; Dominey-Howes *et al.* 2010; Papathoma-Köhle *et al.* 2019) originally designed for tsunami hazards is a representative indicator-based method. Based on PTVA, a similar approach was used for hydrogeomorphic hazards in France (Kappes *et al.* 2012), Peru (Thouret *et al.* 2014; Ettinger *et al.* 2016) and India

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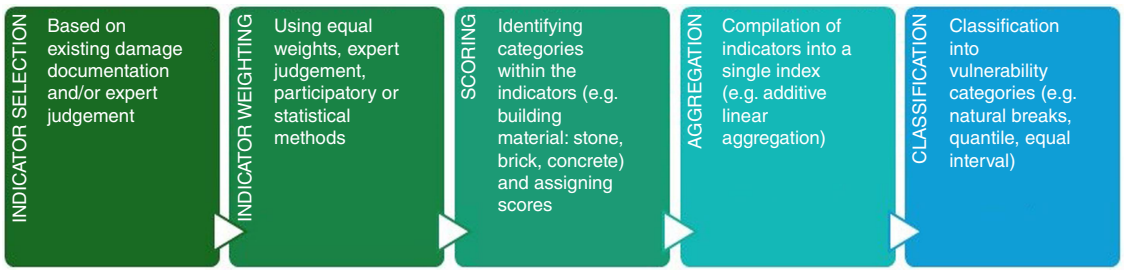


Fig. 3. The stepwise development of a vulnerability index.

(Thennavan *et al.* 2016). The most recent development is a Physical Vulnerability Index (PVI) for dynamic flooding in the European Alps (Papathoma-Köhle *et al.* 2019) and an index for wildfire vulnerability based on empirical data from a wildfire event in Greece in 2018 (Papathoma-Köhle *et al.* 2022*b*). The steps in the development of a vulnerability index are shown in Fig. 3. Following the selection of the relevant indicators, the indicators need to be weighted (in this case study using AHP) and then normalised and aggregated into a single index. The index can be calculated as:

$$\mathrm{PVI} = \sum_{i=1}^{n} w_i I_i$$

In the above equation, w_i expresses the weighting (Table 3) of the scores for each indicator I_i that receives a value of 1, 2, 3, 4 or 5 according to how the specific characteristic affects the vulnerability of the building. In the present study, we assumed that the three categories are equally important for the vulnerability of the building. The reason for assigning equal weights is mainly the fact that these three categories express three different aspects: (a) vulnerability related to whether and at which intensity the fire approaches the settlements where the building is located (*surroundings*); (b) vulnerability related to the fire reaching the building (*exterior of the building*); and finally, (c) the way the building reacts to the fire (*building characteristics*).

The assessment of each building can be done in a spreadsheet in the form of an assessment tool, as shown in Fig. 4. In this figure, the scoring for each indicator can be found, ranging from 1 to 5, depending on whether it reduces vulnerability or not. The scale of 1–5 was chosen to capture the variation of each indicator within the area under investigation. Some of the indicators have five scoring classes to choose from (e.g. building material), others have fewer (e.g. roof shape) and others can be scored either 1 or 5 (present or not present), e.g. water sources, buffer strips or firebreaks in the adjacent forest.

This scoring can be modified according to the specifics of each study area to reflect local conditions. Alternatively, the scoring can be based on the judgement of local experts who are familiar with local conditions. The weighting used for the index in this study is based on the national experts, as they have knowledge of the Austrian context. The weighting of the international experts is important because of their long experience with wildfire events, and it is used to validate the weighting of the local stakeholders.

The final step of the vulnerability assessment process is the classification of the vulnerability classes. There are many different methods, depending on the user and the purpose of the study (Papathoma-Köhle *et al.* 2019), including equal intervals, quantiles and natural breaks. It is clear that the choice of classification can guide different actions to reduce vulnerability.

Results and discussion

Despite their different backgrounds and experience with wildfires, the two teams of experts came to very similar conclusions (Fig. 5 and Table 3). For both teams, the presence of protective strips or fire breaks in the nearby forest, the distance of tree crowns to the building and the roof material were the most important features. However, in terms of building characteristics, the Austrian experts considered the roof overhang to be much more important than the international experts. This may be due to local characteristics. Specifically, in some regions of Austria, buildings have wooden roofs with large overhangs. In this case, it may be easier for the roof to burn. In buildings with concrete roofs, the overhang may not be as important. Furthermore, regarding the building environment, the international experts, in contrast to their Austrian counterparts, considered the type of forest more important than the distance to neighbouring buildings. Finally, regarding the building exterior, the distance to the neighbouring forest was the most important indicator for the international experts in contrast to the type of forest, which seemed to be the most important indicator in this category for the Austrian stakeholders.

Investigating the aggregated individual judgements of the two groups of experts allows one to analyse and compare the average pairwise comparisons between the two groups (Fig. 6*a* and *b*). Specifically, investigating differences (Δ) in the weighting of the two groups (Fig. 6*c*) allows us to unearth patterns of agreement or disagreement between them. Whereas the category 'building exterior' shows

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Basis data - please fill in	
Name of investigator	…
Date of the assessment:	DD.MM.YYYY
Address:	…
Building type	-
Living area [m²]:	…
Plot area [m²]:	…

		Indicators	Low vulnerability		Vulnerability		High vulnerability		
			1	2	3	4	5	Assessment	Notes
Vulnerability		**Building environment**	1	2	3	4	5	Score	
	1	Building density	Single building	Semidetached house	Terraced house	-	Housing estate (consisting of detached houses, semi-detached houses, terraced houses or residential complex)		
	2	Distance to neighbouring buildings	>10 m	5–10 m	2–5 m	-	<2 m		
	3	Water sources	Present	-	-	-	Not present		
	4	Vegetation density	Single trees	-	Tree group	Light tree population	Closed tree population		
	5	Forest type	Deciduous trees	Young forest, mixed deciduous forest, mixed coniferous forest	Coniferous trees (excl. spruce/pine)	-	Spruce/pine forest		
	6	Protective strips or fire breaks in adjacent forest	Present	-	-	-	Not present		
		Building exterior	1	2	3	4	5	Score	
	7	Flammable materials (dead wood) and/or objects (garden furniture, etc.)	Not present	-	-	-	Present		
	8	Terrace/floor covering	Stone, concrete, metal	-	Plastic / WPC	-	Wood		
	9	Property boundary (material)	Stone, concrete, metal	-	Plastic	-	Wood		
	10	Terrain slope	<10°	10°-20°	-	20°-30°	>30°		
	11	Type of vegetation on the property	No plants	-	-	Pot plants	Trees, shrubs, high grass		
	12	Distance of vegetation/tree crowns to the building	No vegetation	-	Vegetation >5 m from the building	-	Vegetation <5 m from building/roof		
	13	Distance from forest edge to building	>200 m	100–200 m	-	50–100 m	<50 m		
		Building characteristics	1	2	3	4	5	Score	
	14	Number of floors	1 floor	2 floors	-	-	More than 2 floors		
	15	Building material	Brick, concrete (solid construction)	Metal construction	-	-	Wood, container; (timber frame construction, prefabricated house)		
	16	Facade/cladding	Stone, brick, concrete	Metal	-	-	Wood, insulation		
	17	Roof material	Concrete	Metal, ceramic	-	Wooden construction with ceramic tiles	Wood		
	18	Roof shape (complexity)	Simple construction: flat roof, single-pitch roof	-	Gable roof, hip roof, mansard roof	-	Complex construction: gable roof extended, saddle hipped roof, schedule roof, butterfly roof, parallel roof		
	19	Roof overhang	No overhang	-	Small ovehang <0,5 m	-	Large overhang >0,5 m		
	20	Shutters/roller shutters	Not present	Aluminium	-	Plastic	Wood		
	21	Door/window material	Aluminium	-	Plastic	-	Wood		

Fig. 4. Tool for the assessment of physical vulnerability to wildfire.

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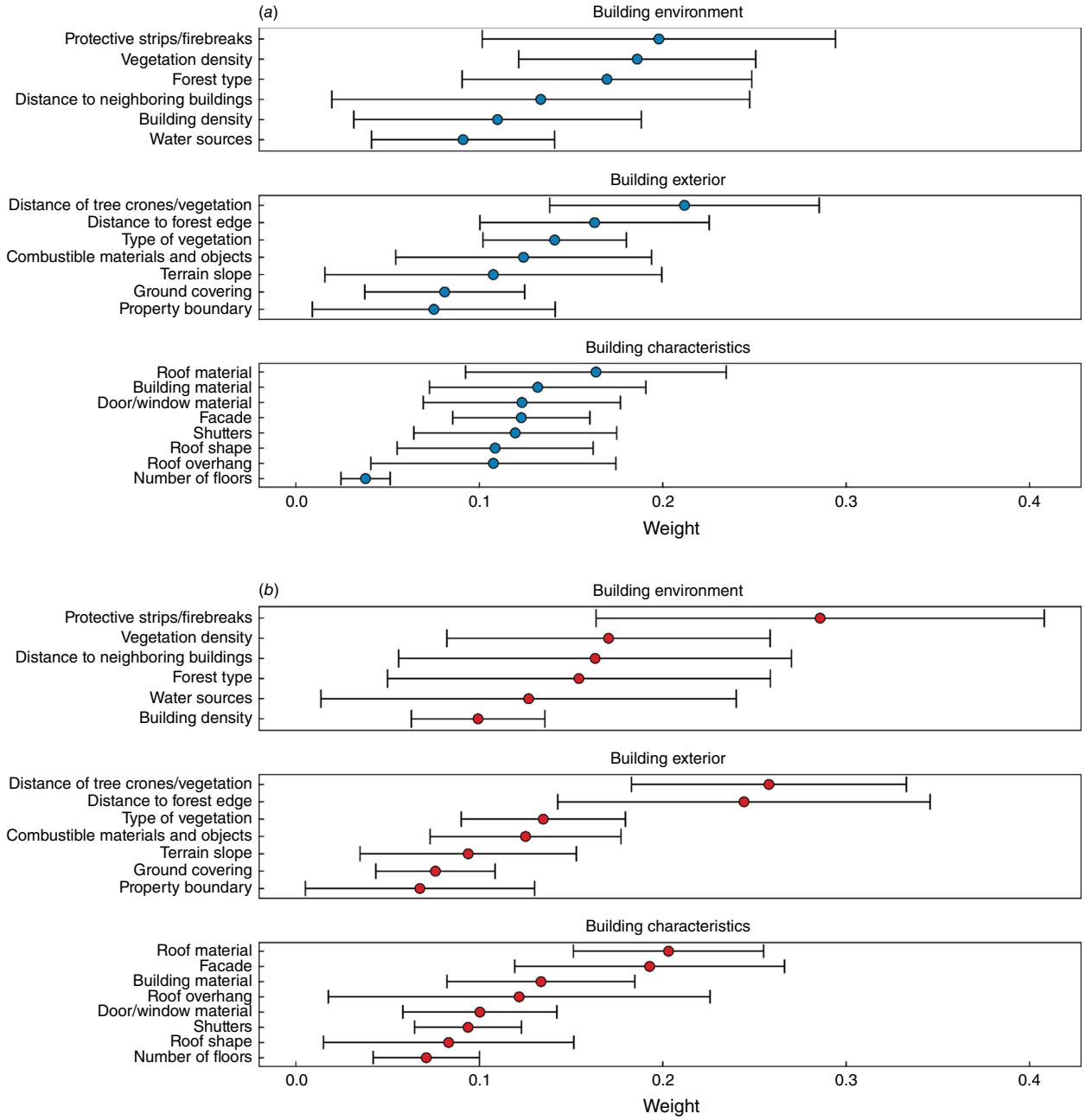


Fig. 5. Weighting of the indicators based on the international experts (*a*), and the Austrian stakeholders (*b*). Dots indicate the weights derived using the geometric mean as an aggregation function. Error bars indicate the standard deviation of the aggregated weights.

comparable judgements between the two groups, major differences can be observed within the categories 'building environment' and 'building characteristics', where Δ is further away from zero. More specifically, Austrian stakeholders consider the presence of water resources to be slightly more important than vegetation density, whereas international experts have the opposite opinion, with a rather pronounced judgement favouring vegetation density over the presence of water resources. This can be explained by the lack of experience with wildfires and the sufficient availability of water in Austria (given the scale of wildfires so far), and the greater water scarcity in countries with regular large-scale wildfires. Furthermore, the presence of vegetation and fuel is directly correlated with the probability of fire occurrence and the potential for fire spread (e.g. Alexandre *et al.* 2016*a*, 2016*b*). Consequently, the experts from countries with a higher incidence of wildfires may have assigned greater importance to vegetation density as

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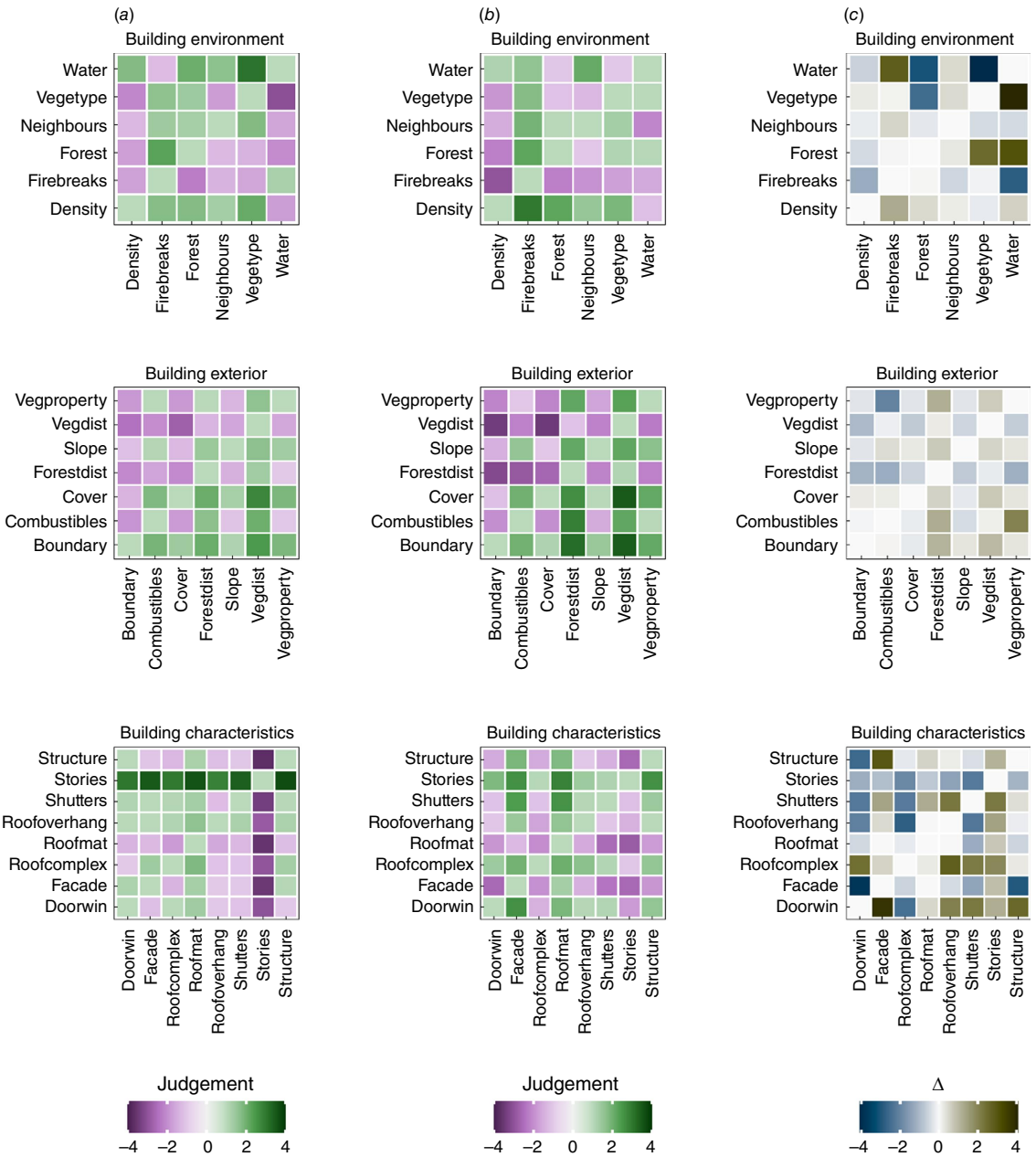


Fig. 6. Aggregated individual judgements of the international experts (*a*), the Austrian stakeholders (*b*), and a demonstration of the differences among the two groups (*c*). Positive values indicate preference of the indicator on the *x*-axis while negative values indicate preference of the indicator on the *y*-axis. Δ is derived as the difference between the international and the Austrian group. See Table 3 for attribute characteristics and definitions of terms.

a risk factor. Moreover, the Austrian group considers the façade more important than the material of the windows and doors, whereas the international experts conclude the opposite. This can be explained by the predominant architecture in each country, with different types of cladding materials in Austria (including wooden shingles) and more homogeneous façade types in other countries. Nevertheless, as in different geographical regions some construction materials are more common than others (Syphard *et al.* 2017; Syphard and Keeley 2019), this indicator is highly case-sensitive. In contrast to the international experts, Austrian stakeholders rate forest type as less important than both the presence of water and vegetation density. This is surprising, given that most wildfires in Austria occur in pine forests, which are not a predominant forest type. Finally, the façade is also considered more important than the building material for the Austrians, whereas the international experts provide the opposite judgement. This can also be attributed

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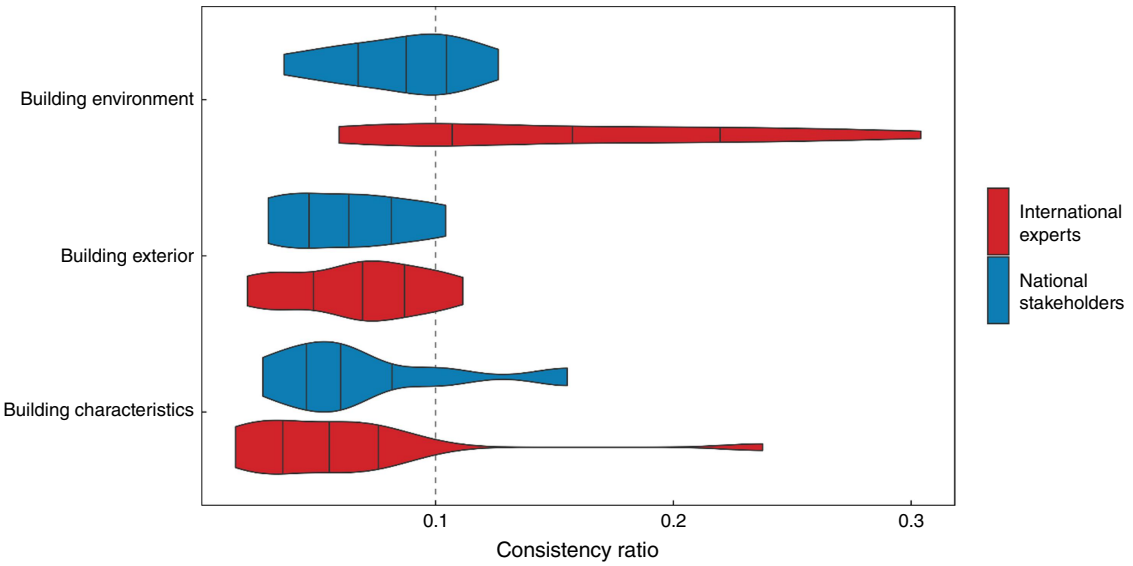


Fig. 7. Distribution of the consistency ratio of all responses for both the international experts and the Austrian stakeholders. Responses with a consistency ratio larger than 0.1 are deemed to be inconsistent according to Saaty (1987).

to the locally predominant architecture, with brick, stone and concrete being more widespread in Austria.

The consistency of the AHP results was assessed (Fig. 7). According to Saaty (1987), responses with a consistency ratio greater than 0.1 are rated inconsistent. Although most of the responses of both groups of experts were consistent, the inconsistency that appeared in the cluster 'building environment' is unexpected. This cluster showed the highest inconsistency (especially from the international experts) despite containing the lowest number of indicators. This inconsistency may reflect the perceived complexity of the cause-effect relationships in this context, which poses a challenge for a consistent ranking of indicators. The higher consistency rate of some responses in the building characteristics group is probably due to the large number of attributes in this class, which inherently increases the risk of inconsistent responses. For this reason, in the future, we can either reduce the number of indicators, inform the decision makers about inconsistencies and ask them to revise their pairwise comparison (Tavana *et al.* 2023), or try to remove inconsistencies computationally (Cao *et al.* 2008).

The questionnaires sent to the experts also included open questions, giving them the opportunity to either question an indicator or suggest a new one. Their responses questioned the relevance of some indicators (e.g. number of storeys, roof overhang, water sources and terrain slope, roof shape). However, some additional indicators were suggested including distance from roads, the defensive potential of the building (firefighting infrastructure), leeward land use, type and cleanliness of the gutters, vegetation connectivity, vegetation maintenance, and maintenance level and the overall condition of the building and surrounding vegetation. Furthermore, some experts suggested indicators related to neighbouring buildings, details of windows and shutters, concavities in the building envelope and the presence of a slab separating the roof from the rest of the building. Some experts also discussed the inclusion of information on preparedness (ability to defend the building, existence of preventive measures) but also on the response of the homeowners (e.g. presence of people in the building). Finally, one of the experts noted that the intensity of the wildfire on each building should also be taken into account as well as the prevailing winds (direction and speed). However, the majority of the responses were in line with the existing indicators and the additional ones suggested will need to be discussed with the local stakeholders in further development of the study to see how relevant they are to the Austrian context (through additional workshops and interviews). In more detail, a summary of the responses of the experts to the open questions can be found in Table 4.

The assessment tool, including all the indicators and the scoring (Fig. 4), was tested on 23 buildings along the Gladiolenweg in the village of Oberschütt (Carinthia, Austria), located in the WUI on the lower southern slope of the Dobratsch mountain. Classification of the vulnerability classes was based on division into equal intervals. It becomes obvious that not all the buildings in the settlement under investigation are equally vulnerable (Fig. 8), demonstrating the spatial variability of the index. Buildings closer to the edge of the forest are more vulnerable, but not always (there are buildings very close to the edge of the forest with moderate vulnerability). Local authorities or individual homeowners can use this information to recommend or implement local adaptation measures (changes to the

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Table 4. Non-relevant and additional indicators proposed by international experts and Austrian stakeholders. The number of experts supporting each indicator is shown in the parentheses.

International experts	
Not relevant	**Additional**
Slope (1)	Distance to roads (1)
Roof overhang (1) (unless design is the focus)	Fire characteristics and fire suppression infrastructure (1)
Number of floors (2)	Absence/presence of people (1)
Roof complexity (1)	Potential for the building to be defended (1)
Water sources (1)	Upwind land use (1)
	Preventive measures and self-protection systems (2)
	Data on recent wildfire events (1)
	Ember screens on the openable portion of the window (1)
	Shutter material (1)
	Type of window glass (1)
	Framing used in the roof space (1)
	Concavities or breakages in the building envelope (2)
	Presence of exterior combustible elements (2)
	Subfloor area used for the storage of combustible material (1)
	Wind speed and direction (1)
	Deck attached to the building (1)
	Fence near the building (1)
	Building (1) and vegetation maintenance (1)
	Fuel moisture content (1)
	Gutter type (material) and cleanliness from leaf debris (1)
	Existence of an inert plate between the roof and the interior of the building (1)
	Condition of vegetation and existence of combustible materials or abandoned cars in the neighbouring property (1)
	Connectivity of vegetation (1)
Austrian stakeholders	
Not relevant	**Additional**
Roof overhang (1)	Orientation of the slope where the building is located (1)
	Altitude (geographical position) (1)
	Ground humidity (1)
	Population density in WUI (1)
	Temporal storage of material between the fire and the building (1)
	Infrastructure (gas tanks, auxiliary buildings) between the fire and the building (1)
	Response time and equipment of the fire brigade (1)
	Exposure of the building to the prevailing wind (1)

building) or other activities (pruning of nearby trees) to increase the resilience of specific buildings. The community can also use this information to plan evacuation routes and prioritise actions. The spatial pattern of vulnerability could also be overlaid with information on the location of vulnerable groups.

Ideally, the relevance and weighting of indicators can be validated using statistical methods (e.g. Boruta selection) after a future disaster (Papathoma-Köhle *et al.* 2022*b*). The relevance of the indicators in question can be checked and additional indicators can be included.

The weighting of the indicators presented here is based on the expert judgement of Austrian stakeholders. It is noteworthy that the weighting of the international experts was not much different. However, if we compare the results with the index developed for Greece using empirical damage data (Papathoma-Köhle *et al.* 2022*b*), we see that in some cases, the results are similar (roof material is the most important indicator for building characteristics in both studies) but in some cases, they are not. For example, although terrain slope is the third most important indicator in the Greek study, it is not particularly important in the present study. However, it should be noted that in the Greek case study, the indicators were not divided into categories. The comparison of the two studies, however, has highlighted the challenges in terms of the transferability of results between countries, given the differences in construction design. It is clear that it is challenging to develop a 'universal model' using expert judgement, but statistical analysis based on damage data may provide more reliable results on the relevance and weighting of indicators. More research is therefore needed on the transferability of such indices.

The Austrian stakeholders who participated in this study were national or regional stakeholders and as such, they understand the Austrian status quo. They may, however, lack local knowledge of particularities and special needs at the municipality level. It would be interesting in the future to see if local stakeholders come to similar conclusions.

Conclusion

The development of a wildfire vulnerability index based on expert judgement is presented with a special focus on the Austrian context. Vulnerability indicators were divided into three categories and then weighted by two different groups of experts using AHP. The results of the weighting showed many similarities between the groups, but also differences. This is the very first attempt to develop such a tool for a country with a poor record of wildfire events and little experience with the phenomenon. Nevertheless, the role of such a tool may be crucial in the near future, as wildfire events are expected to increase in the European

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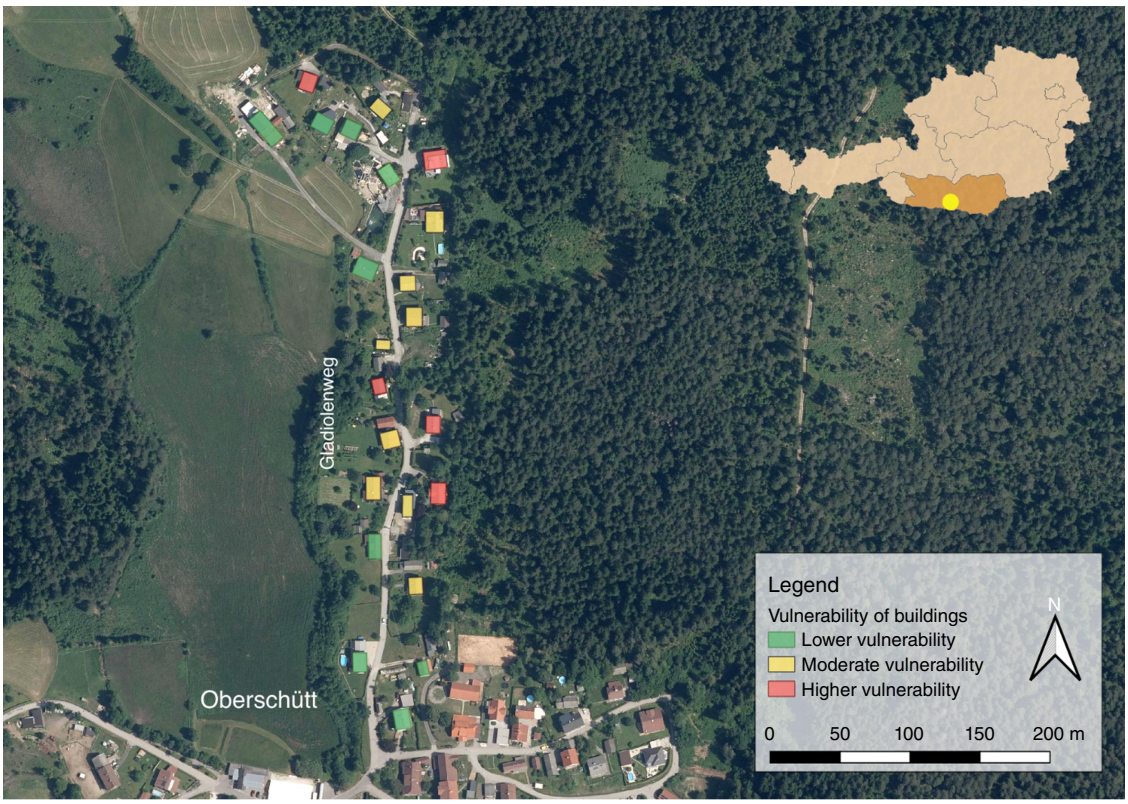


Fig. 8. Implementation of the vulnerability index in an Austrian WUI (village of Oberschütt, Carinthia, Austria - the exact location (yellow dot) in Austria and in Carinthia can be seen in the inset map).

Alps. The results are therefore of interest to other countries facing similar challenges. In this case, local decision-makers will need methods and tools that allow them to focus their resources more efficiently. Given the scarcity of similar studies in the specific geographical area, there is still room for future research, including improving the index to include information on the expected fire and local factors that contribute to its propagation (e.g. prevailing winds), but also local features of buildings, landscape and spatial planning.

Supplementary material

Supplementary material is available online.

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Data availability. Data used to generate the results in the paper are available on request from the corresponding author (anonymously). The questionnaire can be found on GitLab at https://gitlab.com/Rexthor/wildfire-indicators-ahp and as supplementary material The code used for data preparation, data processing, statistical analysis and graphical visualisation of results is available on GitLab at https://gitlab.com/Rexthor/wildfire-indicators-ahp.

Conflicts of interest. The authors declare that they have no conflicts of interest.

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